

Segmentation Benchmarking on Xenium Spatial Transcriptomics

Complete Results — All Tables

Dataset: Xenium FFPE Human Breast (Janesick et al. 2023)

ROI: 2 mm × 2 mm • ~23,600 cells • ~3.4 M transcripts • 380-gene panel

17 segmentation methods: 4 nuclear detectors × 2 expansion strategies
+ Baysor prior confidence sweep + reference-free negative marker validation

Section 1

Dataset & Methods Overview

Table 1.1 — Dataset Summary

Property	Value
Tissue	FFPE Human Breast, invasive ductal carcinoma
Source	Janesick et al. 2023, Nature Communications
Platform	10x Xenium (Custom Add-on Panel)
ROI Dimensions	2 mm × 2 mm
Total Cells (10x native)	23,629
Total Transcripts (ROI)	3,392,051
Gene Panel	380 genes
Companion scRNA-seq	GEO GSE243275

Table 1.2 — Methods Inventory (16 methods)

Method	Family	Detector	Expansion	PSC
10x native	Reference	10x Ranger	Proprietary	—
CellPose	Nuclear	CellPose 3.x	None (nuclear only)	—
StarDist	Nuclear	StarDist 2D_versatile_fluo	None (nuclear only)	—
Mesmer	Nuclear	DeepCell/Mesmer (Docker)	None (nuclear only)	—
10x Ranger	Nuclear	10x Ranger (nuclei)	None (nuclear only)	—
Voronoi (CP)	Voronoi	CellPose	Voronoi (nearest centroid)	—
Voronoi (SD)	Voronoi	StarDist	Voronoi (nearest centroid)	—
Voronoi (M)	Voronoi	Mesmer	Voronoi (nearest centroid)	—
Voronoi (10x)	Voronoi	10x Ranger	Voronoi (nearest centroid)	—
Baysor	Transcript-density	—	Baysor transcript-density EM	0.0
Baysor (CP prior 0.2)	Transcript-density	CellPose	Baysor + nuclear prior	0.2
Baysor (CP prior 0.8)	Transcript-density	CellPose	Baysor + nuclear prior	0.8
Baysor (CP prior 1.0)	Transcript-density	CellPose	Baysor + nuclear prior	1.0
Baysor (SD prior 1.0)	Transcript-density	StarDist	Baysor + nuclear prior	1.0
Baysor (M prior 1.0)	Transcript-density	Mesmer	Baysor + nuclear prior	1.0
Baysor (10x prior 1.0)	Transcript-density	10x Ranger	Baysor + nuclear prior	1.0

Table 1.3 — Cluster Annotations (10x Native Reference)

Leiden Cluster	Cell Type	n_cells
0	Luminal epithelial	2,799
1	Luminal epithelial	2,926
2	Macrophages	335
3	Luminal epithelial	902
4	Myoepithelial	1,025
5	T cells	2,861
6	B cells	758
7	Macrophages	2,612
8	Luminal epithelial	1,921
9	CAFs	3,314
10	Smooth muscle	870
11	Endothelial	1,508
12	Plasma cells	266
13	CAFs	1,333
14	Adipocytes	199

Section 2

Cell & Transcript Recovery

Table 2.1 — Cell Counts & Transcript Recovery (All Methods)

Method	Family	n_cells	Total Transcripts	Median tx/cell	Mean tx/cell	Capture Rate
10x native	Reference	23,629	3,358,054	124	142.12	99.0%
CellPose	Nuclear	20,166	1,202,133	49	59.61	35.4%
StarDist	Nuclear	24,745	1,385,547	45	55.99	40.8%
Mesmer	Nuclear	21,697	1,757,936	70	81.02	51.8%
10x Ranger	Nuclear	23,624	1,290,456	45	54.62	38.0%
Voronoi (CP)	Voronoi	20,166	3,392,051	149	168.21	100.0%
Voronoi (SD)	Voronoi	24,743	3,392,051	122	137.09	100.0%
Voronoi (M)	Voronoi	21,697	3,392,051	142	156.34	100.0%
Voronoi (10x)	Voronoi	23,622	3,392,051	128	143.60	100.0%
Baysor	Transcript-density	18,321	3,344,675	53	182.56	98.6%
Baysor (CP prior 0.2)	Transcript-density	19,061	3,346,292	53	175.56	98.7%
Baysor (CP prior 0.8)	Transcript-density	29,771	3,352,843	67	112.62	98.8%
Baysor (CP prior 1.0)	Transcript-density	30,473	3,353,065	69	110.03	98.9%
Baysor (SD prior 1.0)	Transcript-density	34,230	3,354,017	63	97.98	98.9%
Baysor (M prior 1.0)	Transcript-density	31,764	3,355,569	74	105.64	98.9%
Baysor (10x prior 1.0)	Transcript-density	33,113	3,354,047	65	101.29	98.9%

Table 2.2 — Cell Counts & Transcript Recovery (1 mm² Subset)

Method	n_cells	Total Transcripts	Median tx/cell	Mean tx/cell	Capture Rate
10x native	5,239	756,028	126	144.31	98.1%
CellPose	4,459	265,304	48	59.50	34.4%
StarDist	5,428	305,576	44	56.30	39.6%
Baysor	4,514	759,617	50	168.28	98.6%

Table 2.3 — Cell/Nuclear Size Distributions

Method	Size Metric	P10	P25	Median	P75	P90	Mean	Std
10x native	n_transcripts	52.00	80.00	124.00	184.00	255.20	142.12	86.23
CellPose	area	246.00	401.00	653.00	1055.00	1495.00	783.28	518.91
StarDist	area	220.00	369.00	602.00	1008.00	1481.00	748.89	527.06
Mesmer	area	391.60	656.00	1021.00	1519.00	2042.00	1142.42	665.89
10x Ranger	area	222.00	366.00	592.00	968.00	1389.00	722.32	490.00
Voronoi (CP)	area	246.00	401.00	653.00	1055.00	1495.00	783.28	518.91
Voronoi (SD)	area	220.00	369.00	602.00	1008.00	1481.00	748.94	527.04
Voronoi (M)	area	391.60	656.00	1021.00	1519.00	2042.00	1142.42	665.89
Voronoi (10x)	area	223.00	366.00	592.00	968.00	1389.00	722.37	489.99
Baysor	n_transcripts	4.00	12.00	53.00	173.00	465.00	182.56	357.14
Baysor (CP prior 0.2)	n_transcripts	5.00	13.00	53.00	164.00	439.00	175.56	342.50
Baysor (CP prior 0.8)	n_transcripts	7.00	23.00	67.00	138.00	279.00	112.62	143.11
Baysor (CP prior 1.0)	n_transcripts	6.00	23.00	69.00	140.00	265.00	110.03	132.63
Baysor (SD prior 1.0)	n_transcripts	6.00	22.00	63.00	128.00	232.00	97.98	114.67
Baysor (M prior 1.0)	n_transcripts	5.00	22.00	74.00	145.00	245.00	105.64	114.80
Baysor (10x prior 1.0)	n_transcripts	6.00	22.00	65.00	130.00	240.00	101.29	120.16

Size metric is 'area' (pixels²) for morphological methods, 'n_transcripts' for Baysor/10x native.

Table 2.4 — Cell/Nuclear Size Distributions (1 mm² Subset)

Method	Size Metric	P10	P25	Median	P75	P90	Mean	Std
10x native	n_transcripts	52.00	81.00	126.00	189.00	260.00	144.31	86.40
CellPose	area	235.00	385.50	619.00	1076.50	1587.40	788.79	561.97
StarDist	area	214.70	360.00	576.00	1005.00	1589.30	755.47	568.01
Baysor	n_transcripts	5.00	12.00	50.00	159.00	409.00	168.28	334.28

Section 3

Per-Cell Expression Correlation

Table 3.1 — Per-Cell Expression Correlation vs. 10x Native

Method	n_pairs	Median r	P25 r	P75 r
CellPose	18,944	0.822	0.680	0.900
StarDist	21,409	0.826	0.694	0.904
Mesmer	20,591	0.879	0.769	0.937
10x Ranger	23,128	0.822	0.685	0.901
Voronoi (CP)	18,966	0.959	0.917	0.979
Voronoi (SD)	21,427	0.959	0.919	0.979
Voronoi (M)	20,595	0.964	0.930	0.981
Voronoi (10x)	0	nan	nan	nan
Baysor	10,953	0.786	0.518	0.914
Baysor (CP prior 0.2)	11,454	0.798	0.536	0.918
Baysor (CP prior 0.8)	20,108	0.884	0.740	0.945
Baysor (CP prior 1.0)	20,308	0.902	0.773	0.954
Baysor (SD prior 1.0)	21,814	0.905	0.783	0.955
Baysor (M prior 1.0)	21,148	0.924	0.819	0.965
Baysor (10x prior 1.0)	22,910	0.914	0.795	0.960

Table 3.2 — Expression Correlation (Non-10x Reference Pairs)

Comparison	n_pairs	Median r	P25 r	P75 r
CellPose → Baysor (original)	8,928	0.731	0.532	0.832
cellpose → 10x	18,944	0.822	0.680	0.900
cellpose → baysor → prior	9,555	0.742	0.568	0.837
cellpose → stardist	19,431	0.964	0.920	0.984

Section 4

Clustering Agreement vs. 10x Native

Table 4.1 — Master Agreement Summary (All Comparisons vs. 10x Native)

Method	Matched Pairs	Median r	ARI	Hung. Disagree	Hung. Moran's I	Argmax Disagree	Argmax Moran's I
CellPose	18,966	0.822	0.547	30.8%	0.178	30.5%	0.189
StarDist	21,429	0.826	0.544	33.5%	0.215	33.4%	0.221
Mesmer	20,595	0.879	0.557	27.9%	0.090	27.3%	0.098
10x Ranger	23,155	0.822		35.0%	0.191	34.5%	0.203
Voronoi (CP)	18,966	0.959	0.629	21.9%	0.076	21.9%	0.076
Voronoi (SD)	21,428	0.959	0.584	31.9%	0.194	27.7%	0.229
Voronoi (M)	20,595	0.964	0.686	18.8%	0.161	18.8%	0.161
Voronoi (10x)	23,153	nan		28.3%	0.172	25.8%	0.168
Baysor	10,953	0.786	0.305	51.7%	0.033	43.8%	0.079
Baysor (CP prior 0.2)	11,454	0.798		51.9%	0.036	39.2%	0.086
Baysor (CP prior 0.8)	20,108	0.884		38.8%	0.219	37.4%	0.234
Baysor (CP prior 1.0)	20,308	0.902		33.8%	0.111	32.1%	0.122
Baysor (SD prior 1.0)	21,814	0.905		37.7%	0.136	32.9%	0.170
Baysor (M prior 1.0)	21,148	0.924		32.3%	0.115	30.7%	0.119
Baysor (10x prior 1.0)	22,910	0.914		34.7%	0.208	33.1%	0.204

Table 4.2 — Hungarian vs. Argmax Matching Accuracy

Comparison	Total Pairs	Hungarian Accuracy	Many-to-One Accuracy	Gap
CellPose	18,966	0.692	0.695	0.003
StarDist	21,429	0.665	0.666	0.001
Mesmer	20,595	0.721	0.727	0.005
Voronoi (CP)	18,966	0.781	0.781	0.000
Voronoi (M)	20,595	0.812	0.812	0.000
Baysor	10,953	0.483	0.562	0.080

Table 4.3 — Pairwise ARI (All Method Pairs)

Method A	Method B	ARI	n_matched
10x native	CellPose	0.547	18,966
10x native	StarDist	0.544	21,429
10x native	Mesmer	0.557	20,595
10x native	Voronoi (CP)	0.629	18,966
10x native	Voronoi (SD)	0.584	21,428
10x native	Voronoi (M)	0.686	20,595
10x native	Baysor	0.305	10,953
CellPose	StarDist	0.764	19,460
CellPose	Mesmer	0.606	18,762
CellPose	Voronoi (CP)	0.533	20,166
CellPose	Voronoi (SD)	0.557	19,460
CellPose	Voronoi (M)	0.556	18,762
CellPose	Baysor	0.415	8,947
StarDist	Mesmer	0.638	20,941
StarDist	Voronoi (CP)	0.525	19,460
StarDist	Voronoi (SD)	0.568	24,743
StarDist	Voronoi (M)	0.540	20,941
StarDist	Baysor	0.411	9,826
Mesmer	Voronoi (CP)	0.480	18,762
Mesmer	Voronoi (SD)	0.535	20,941
Mesmer	Voronoi (M)	0.527	21,697
Mesmer	Baysor	0.455	9,696
Voronoi (CP)	Voronoi (SD)	0.581	19,460
Voronoi (CP)	Voronoi (M)	0.661	18,762
Voronoi (CP)	Baysor	0.336	8,947
Voronoi (SD)	Voronoi (M)	0.633	20,941
Voronoi (SD)	Baysor	0.361	9,826
Voronoi (M)	Baysor	0.377	9,696

Table 4.4 — Pairwise ARI Matrix (8 Core Methods)

Method	10x native	CellPose	StarDist	Mesmer	Voronoi (CP)	Voronoi (SD)	Voronoi (M)	Baysor
10x native	1.000	0.547	0.545	0.557	0.630	0.584	0.686	0.305
CellPose	0.547	1.000	0.764	0.606	0.533	0.557	0.556	0.415
StarDist	0.545	0.764	1.000	0.639	0.525	0.568	0.540	0.411
Mesmer	0.557	0.606	0.639	1.000	0.480	0.535	0.528	0.455
Voronoi (CP)	0.630	0.533	0.525	0.480	1.000	0.581	0.661	0.336
Voronoi (SD)	0.584	0.557	0.568	0.535	0.581	1.000	0.633	0.361
Voronoi (M)	0.686	0.556	0.540	0.528	0.661	0.633	1.000	0.377
Baysor	0.305	0.415	0.411	0.455	0.336	0.361	0.377	1.000

Gap: Only 8 methods in matrix. Missing: Baysor prior variants, 10x Ranger, Voronoi(10x).

Section 5

Resolution Sensitivity

Table 5.1 — Resolution Sensitivity (Hungarian) (page 1/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.3	Baysor	0.392	8	11	36.0%	0.025
0.3	Baysor (CP prior 0.8)	0.653	8	9	23.2%	0.214
0.3	Baysor (CP prior 1.0)	0.650	8	10	24.2%	0.230
0.3	Baysor (M prior 1.0)	0.671	8	10	21.2%	0.200
0.3	Baysor (SD prior 1.0)	0.673	8	9	20.8%	0.206
0.3	CellPose	0.772	8	7	16.1%	0.135
0.3	Mesmer	0.707	8	8	18.5%	0.260
0.3	StarDist	0.693	8	6	21.0%	0.261
0.3	Voronoi (CP)	0.720	8	7	21.4%	0.174
0.3	Voronoi (M)	0.825	8	8	9.3%	0.063
0.3	Voronoi (SD)	0.732	8	7	18.3%	0.146
0.5	Baysor	0.403	9	14	37.6%	0.042
0.5	Baysor (CP prior 0.8)	0.647	9	12	24.4%	0.230
0.5	Baysor (CP prior 1.0)	0.584	9	13	25.8%	0.154
0.5	Baysor (M prior 1.0)	0.723	9	13	18.9%	0.105
0.5	Baysor (SD prior 1.0)	0.655	9	13	24.1%	0.228
0.5	CellPose	0.776	9	9	16.7%	0.137
0.5	Mesmer	0.682	9	10	17.0%	0.191
0.5	StarDist	0.650	9	9	22.2%	0.166
0.5	Voronoi (CP)	0.770	9	9	14.6%	0.211
0.5	Voronoi (M)	0.716	9	9	14.5%	0.232
0.5	Voronoi (SD)	0.798	9	8	12.3%	0.096
0.6	Baysor	0.349	11	14	44.0%	0.049
0.6	Baysor (CP prior 0.8)	0.530	11	15	34.4%	0.210
0.6	Baysor (CP prior 1.0)	0.534	11	14	33.8%	0.220
0.6	Baysor (M prior 1.0)	0.545	11	14	32.9%	0.180
0.6	Baysor (SD prior 1.0)	0.553	11	14	30.6%	0.228
0.6	CellPose	0.567	11	10	33.3%	0.241

Table 5.1 — Resolution Sensitivity (Hungarian) (page 2/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.6	Mesmer	0.617	11	11	27.6%	0.278
0.6	StarDist	0.577	11	10	33.0%	0.317
0.6	Voronoi (CP)	0.666	11	11	20.5%	0.209
0.6	Voronoi (M)	0.756	11	12	12.9%	0.106
0.6	Voronoi (SD)	0.640	11	11	21.2%	0.170
0.7	Baysor	0.288	13	18	53.6%	0.029
0.7	Baysor (CP prior 0.8)	0.509	13	14	38.2%	0.245
0.7	Baysor (CP prior 1.0)	0.513	13	15	36.3%	0.168
0.7	Baysor (M prior 1.0)	0.538	13	17	34.1%	0.182
0.7	Baysor (SD prior 1.0)	0.555	13	15	31.3%	0.231
0.7	CellPose	0.589	13	11	24.9%	0.141
0.7	Mesmer	0.602	13	12	29.4%	0.332
0.7	StarDist	0.579	13	10	31.5%	0.308
0.7	Voronoi (CP)	0.635	13	12	20.7%	0.143
0.7	Voronoi (M)	0.647	13	14	19.9%	0.181
0.7	Voronoi (SD)	0.653	13	13	19.4%	0.179
0.8	Baysor	0.330	13	19	47.3%	0.050
0.8	Baysor (CP prior 0.8)	0.479	13	17	40.6%	0.206
0.8	Baysor (CP prior 1.0)	0.496	13	19	33.9%	0.102
0.8	Baysor (M prior 1.0)	0.491	13	18	41.0%	0.205
0.8	Baysor (SD prior 1.0)	0.555	13	18	26.8%	0.050
0.8	CellPose	0.586	13	11	24.0%	0.129
0.8	Mesmer	0.605	13	13	24.7%	0.176
0.8	StarDist	0.582	13	12	25.9%	0.150
0.8	Voronoi (CP)	0.594	13	13	25.2%	0.165
0.8	Voronoi (M)	0.632	13	14	22.4%	0.177
0.8	Voronoi (SD)	0.584	13	15	29.5%	0.145
0.9	Baysor	0.296	14	21	53.5%	0.025

Table 5.1 — Resolution Sensitivity (Hungarian) (page 3/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.9	Baysor (CP prior 0.8)	0.467	14	19	41.6%	0.215
0.9	Baysor (CP prior 1.0)	0.512	14	19	31.6%	0.081
0.9	Baysor (M prior 1.0)	0.549	14	22	30.9%	0.088
0.9	Baysor (SD prior 1.0)	0.555	14	20	27.2%	0.075
0.9	CellPose	0.589	14	12	24.6%	0.115
0.9	Mesmer	0.599	14	13	25.7%	0.192
0.9	StarDist	0.578	14	12	26.8%	0.167
0.9	Voronoi (CP)	0.588	14	14	25.3%	0.133
0.9	Voronoi (M)	0.636	14	14	21.7%	0.161
0.9	Voronoi (SD)	0.562	14	15	30.7%	0.164
1.0	Baysor	0.305	15	21	51.7%	0.033
1.0	Baysor (CP prior 0.8)	0.488	15	20	38.8%	0.219
1.0	Baysor (CP prior 1.0)	0.501	15	21	33.8%	0.112
1.0	Baysor (M prior 1.0)	0.518	15	23	32.3%	0.115
1.0	Baysor (SD prior 1.0)	0.498	15	22	37.7%	0.136
1.0	CellPose	0.547	15	13	30.8%	0.178
1.0	Mesmer	0.557	15	15	27.9%	0.090
1.0	StarDist	0.544	15	12	33.5%	0.214
1.0	Voronoi (CP)	0.629	15	14	21.9%	0.076
1.0	Voronoi (M)	0.686	15	14	18.8%	0.161
1.0	Voronoi (SD)	0.584	15	16	31.9%	0.194
1.2	Baysor	0.285	16	24	53.4%	0.042
1.2	Baysor (CP prior 0.8)	0.445	16	24	40.8%	0.110
1.2	Baysor (CP prior 1.0)	0.469	16	26	42.0%	0.132
1.2	Baysor (M prior 1.0)	0.485	16	27	38.0%	0.095
1.2	Baysor (SD prior 1.0)	0.437	16	23	43.4%	0.115
1.2	CellPose	0.512	16	14	35.4%	0.166
1.2	Mesmer	0.520	16	15	31.9%	0.087

Table 5.1 — Resolution Sensitivity (Hungarian) (page 4/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
1.2	StarDist	0.483	16	16	38.8%	0.123
1.2	Voronoi (CP)	0.591	16	16	28.6%	0.097
1.2	Voronoi (M)	0.611	16	17	25.8%	0.135
1.2	Voronoi (SD)	0.547	16	16	32.6%	0.138
1.5	Baysor	0.267	22	29	56.0%	0.051
1.5	Baysor (CP prior 0.8)	0.422	22	27	46.2%	0.126
1.5	Baysor (CP prior 1.0)	0.457	22	30	45.3%	0.153
1.5	Baysor (M prior 1.0)	0.458	22	33	41.9%	0.087
1.5	Baysor (SD prior 1.0)	0.438	22	29	46.9%	0.129
1.5	CellPose	0.496	22	19	38.2%	0.094
1.5	Mesmer	0.501	22	17	38.7%	0.097
1.5	StarDist	0.455	22	20	41.4%	0.077
1.5	Voronoi (CP)	0.562	22	19	31.6%	0.096
1.5	Voronoi (M)	0.529	22	21	32.5%	0.099
1.5	Voronoi (SD)	0.581	22	21	29.5%	0.088
2.0	Baysor	0.274	24	39	58.2%	0.024
2.0	Baysor (CP prior 0.8)	0.386	24	35	46.6%	0.043
2.0	Baysor (CP prior 1.0)	0.383	24	36	48.5%	0.068
2.0	Baysor (M prior 1.0)	0.437	24	37	45.3%	0.101
2.0	Baysor (SD prior 1.0)	0.435	24	38	46.5%	0.112
2.0	CellPose	0.402	24	23	46.7%	0.086
2.0	Mesmer	0.471	24	22	39.8%	0.070
2.0	StarDist	0.422	24	24	45.4%	0.070
2.0	Voronoi (CP)	0.461	24	27	37.2%	0.105
2.0	Voronoi (M)	0.559	24	25	29.4%	0.059
2.0	Voronoi (SD)	0.576	24	24	28.8%	0.063

Table 5.2 — Resolution Sensitivity (Argmax) (page 1/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.3	Baysor	0.392	8	11	32.3%	0.040
0.3	Baysor (CP prior 0.8)	0.653	8	9	22.2%	0.221
0.3	Baysor (CP prior 1.0)	0.650	8	10	20.6%	0.229
0.3	Baysor (M prior 1.0)	0.671	8	10	19.7%	0.217
0.3	Baysor (SD prior 1.0)	0.673	8	9	19.7%	0.217
0.3	CellPose	0.772	8	7	14.7%	0.094
0.3	Mesmer	0.707	8	8	15.7%	0.269
0.3	StarDist	0.693	8	6	19.7%	0.247
0.3	Voronoi (CP)	0.720	8	7	21.4%	0.174
0.3	Voronoi (M)	0.825	8	8	9.3%	0.063
0.3	Voronoi (SD)	0.732	8	7	18.3%	0.146
0.5	Baysor	0.403	9	14	33.3%	0.056
0.5	Baysor (CP prior 0.8)	0.647	9	12	22.7%	0.235
0.5	Baysor (CP prior 1.0)	0.584	9	13	22.7%	0.236
0.5	Baysor (M prior 1.0)	0.723	9	13	16.8%	0.106
0.5	Baysor (SD prior 1.0)	0.655	9	13	22.2%	0.235
0.5	CellPose	0.776	9	9	13.2%	0.080
0.5	Mesmer	0.682	9	10	16.5%	0.287
0.5	StarDist	0.650	9	9	18.8%	0.150
0.5	Voronoi (CP)	0.770	9	9	12.2%	0.099
0.5	Voronoi (M)	0.716	9	9	14.5%	0.232
0.5	Voronoi (SD)	0.798	9	8	12.3%	0.096
0.6	Baysor	0.349	11	14	35.4%	0.074
0.6	Baysor (CP prior 0.8)	0.530	11	15	29.8%	0.239
0.6	Baysor (CP prior 1.0)	0.534	11	14	29.3%	0.245
0.6	Baysor (M prior 1.0)	0.545	11	14	31.3%	0.195
0.6	Baysor (SD prior 1.0)	0.553	11	14	29.0%	0.252
0.6	CellPose	0.567	11	10	29.8%	0.261

Table 5.2 — Resolution Sensitivity (Argmax) (page 2/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.6	Mesmer	0.617	11	11	27.3%	0.283
0.6	StarDist	0.577	11	10	29.6%	0.344
0.6	Voronoi (CP)	0.666	11	11	16.1%	0.102
0.6	Voronoi (M)	0.756	11	12	11.1%	0.040
0.6	Voronoi (SD)	0.640	11	11	21.2%	0.171
0.7	Baysor	0.288	13	18	37.1%	0.086
0.7	Baysor (CP prior 0.8)	0.509	13	14	36.7%	0.269
0.7	Baysor (CP prior 1.0)	0.513	13	15	34.6%	0.181
0.7	Baysor (M prior 1.0)	0.538	13	17	32.4%	0.200
0.7	Baysor (SD prior 1.0)	0.555	13	15	29.5%	0.256
0.7	CellPose	0.589	13	11	24.6%	0.178
0.7	Mesmer	0.602	13	12	29.2%	0.340
0.7	StarDist	0.579	13	10	31.2%	0.337
0.7	Voronoi (CP)	0.635	13	12	18.3%	0.070
0.7	Voronoi (M)	0.647	13	14	16.9%	0.169
0.7	Voronoi (SD)	0.653	13	13	19.4%	0.179
0.8	Baysor	0.330	13	19	38.1%	0.091
0.8	Baysor (CP prior 0.8)	0.479	13	17	34.5%	0.258
0.8	Baysor (CP prior 1.0)	0.496	13	19	28.6%	0.106
0.8	Baysor (M prior 1.0)	0.491	13	18	34.3%	0.254
0.8	Baysor (SD prior 1.0)	0.555	13	18	25.0%	0.056
0.8	CellPose	0.586	13	11	24.0%	0.129
0.8	Mesmer	0.605	13	13	24.1%	0.186
0.8	StarDist	0.582	13	12	25.7%	0.150
0.8	Voronoi (CP)	0.594	13	13	20.5%	0.077
0.8	Voronoi (M)	0.632	13	14	19.2%	0.154
0.8	Voronoi (SD)	0.584	13	15	23.7%	0.184
0.9	Baysor	0.296	14	21	40.0%	0.128

Table 5.2 — Resolution Sensitivity (Argmax) (page 3/4)

Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
0.9	Baysor (CP prior 0.8)	0.467	14	19	36.6%	0.283
0.9	Baysor (CP prior 1.0)	0.512	14	19	27.7%	0.091
0.9	Baysor (M prior 1.0)	0.549	14	22	24.2%	0.083
0.9	Baysor (SD prior 1.0)	0.555	14	20	25.4%	0.086
0.9	CellPose	0.589	14	12	24.3%	0.121
0.9	Mesmer	0.599	14	13	25.3%	0.206
0.9	StarDist	0.578	14	12	26.7%	0.173
0.9	Voronoi (CP)	0.588	14	14	22.5%	0.076
0.9	Voronoi (M)	0.636	14	14	21.3%	0.180
0.9	Voronoi (SD)	0.562	14	15	24.2%	0.203
1.0	Baysor	0.305	15	21	43.8%	0.079
1.0	Baysor (CP prior 0.8)	0.488	15	20	37.4%	0.234
1.0	Baysor (CP prior 1.0)	0.501	15	21	32.1%	0.122
1.0	Baysor (M prior 1.0)	0.518	15	23	30.7%	0.119
1.0	Baysor (SD prior 1.0)	0.498	15	22	32.9%	0.170
1.0	CellPose	0.547	15	13	30.5%	0.189
1.0	Mesmer	0.557	15	15	27.4%	0.098
1.0	StarDist	0.544	15	12	33.4%	0.221
1.0	Voronoi (CP)	0.629	15	14	21.9%	0.076
1.0	Voronoi (M)	0.686	15	14	18.8%	0.161
1.0	Voronoi (SD)	0.584	15	16	27.7%	0.229
1.2	Baysor	0.285	16	24	45.4%	0.087
1.2	Baysor (CP prior 0.8)	0.445	16	24	35.8%	0.121
1.2	Baysor (CP prior 1.0)	0.469	16	26	36.6%	0.164
1.2	Baysor (M prior 1.0)	0.485	16	27	32.8%	0.089
1.2	Baysor (SD prior 1.0)	0.437	16	23	38.2%	0.161
1.2	CellPose	0.512	16	14	34.9%	0.175
1.2	Mesmer	0.520	16	15	31.3%	0.090

Table 5.2 — Resolution Sensitivity (Argmax) (page 4/4)

	Resolution	Method	ARI	Clusters (10x)	Clusters (comp)	Disagree %	Moran's I
	1.2	StarDist	0.483	16	16	36.9%	0.116
	1.2	Voronoi (CP)	0.591	16	16	24.5%	0.089
	1.2	Voronoi (M)	0.611	16	17	21.0%	0.117
	1.2	Voronoi (SD)	0.547	16	16	27.7%	0.163
	1.5	Baysor	0.267	22	29	54.1%	0.070
	1.5	Baysor (CP prior 0.8)	0.422	22	27	45.3%	0.132
	1.5	Baysor (CP prior 1.0)	0.457	22	30	44.2%	0.165
	1.5	Baysor (M prior 1.0)	0.458	22	33	40.1%	0.087
	1.5	Baysor (SD prior 1.0)	0.438	22	29	45.0%	0.135
	1.5	CellPose	0.496	22	19	37.9%	0.092
	1.5	Mesmer	0.501	22	17	37.9%	0.100
	1.5	StarDist	0.455	22	20	41.0%	0.082
	1.5	Voronoi (CP)	0.562	22	19	31.2%	0.094
	1.5	Voronoi (M)	0.529	22	21	31.2%	0.102
	1.5	Voronoi (SD)	0.581	22	21	28.8%	0.089
	2.0	Baysor	0.274	24	39	52.5%	0.059
	2.0	Baysor (CP prior 0.8)	0.386	24	35	45.1%	0.043
	2.0	Baysor (CP prior 1.0)	0.383	24	36	45.3%	0.088
	2.0	Baysor (M prior 1.0)	0.437	24	37	42.3%	0.119
	2.0	Baysor (SD prior 1.0)	0.435	24	38	43.9%	0.111
	2.0	CellPose	0.402	24	23	41.0%	0.084
	2.0	Mesmer	0.471	24	22	38.2%	0.092
	2.0	StarDist	0.422	24	24	42.5%	0.080
	2.0	Voronoi (CP)	0.461	24	27	31.1%	0.073
	2.0	Voronoi (M)	0.559	24	25	26.2%	0.052
	2.0	Voronoi (SD)	0.576	24	24	28.1%	0.067

Gap: 11 of 17 methods tested. Missing: 10x Ranger, Voronoi(10x), Baysor(CP prior 0.2), Baysor(10x prior 1.0).

Section 6

Confusion Matrices (Cluster-Level)

Table 6.1 — Confusion Matrix: 10x Native vs. 10x Ranger

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12
0: Luminal epithelial	950	1715	3	2	1	72	0	0	4	31	3	2	1
1: Luminal epithelial	1246	1465	1	0	3	128	0	0	2	21	2	1	0
2: Macrophages	18	0	202	35	1	4	1	12	10	39	6	2	0
3: Luminal epithelial	564	79	1	9	5	177	0	9	7	43	1	1	1
4: Myoepithelial	163	140	16	4	572	60	1	8	24	27	2	6	0
5: T cells	2	0	116	2501	0	0	18	63	19	52	24	6	5
6: B cells	0	0	37	33	1	0	599	16	13	29	8	8	2
7: Macrophages	0	0	2111	76	2	0	20	109	58	97	42	8	5
8: Luminal epithelial	123	227	4	1	5	1471	0	5	6	32	5	0	0
9: CAFs	0	0	215	59	7	0	19	2710	50	75	101	10	2
10: Smooth muscle	0	0	21	14	5	0	1	47	681	26	10	30	0
11: Endothelial	4	0	32	51	5	0	5	27	118	12	6	1208	1
12: Plasma cells	0	0	19	18	0	0	9	35	7	4	5	2	161
13: CAFs	14	0	124	35	4	17	10	948	48	66	25	15	0
14: Adipocytes	0	0	27	18	0	0	4	42	41	20	1	17	5

Table 6.2 — Confusion Matrix: 10x Native vs. Baysor

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0: Luminal epithelial	0	3	17	2	1	2	7	4	1	2	15	0	0	7	0	2	1	0	0	0	242
1: Luminal epithelial	2	39	6	7	2	3	6	0	0	1	28	5	4	2	1	4	0	0	0	4	296
2: Macrophages	10	11	143	2	1	10	11	6	1	25	9	0	3	5	3	5	3	0	0	0	0
3: Luminal epithelial	38	132	15	11	6	2	8	2	7	7	10	1	3	4	7	6	2	1	0	2	11
4: Myoepithelial	14	7	16	8	8	7	8	3	0	5	9	20	1	5	2	13	1	1	0	284	33
5: T cells	74	3	74	28	26	23	12	3	8	1158	23	5	48	9	142	26	15	5	10	0	0
6: B cells	19	0	19	9	5	9	4	2	0	191	5	4	10	5	35	5	2	1	5	1	0
7: Macrophages	70	7	687	24	35	516	22	16	11	99	25	4	56	43	136	25	15	3	9	1	0
8: Luminal epithelial	42	306	11	6	8	4	11	3	1	1	21	4	14	0	15	10	7	0	0	3	21
9: CAFs	476	2	111	47	49	89	29	13	6	61	39	5	83	54	1163	64	41	6	8	4	0
10: Smooth muscle	35	1	23	16	16	14	9	2	2	12	10	194	23	7	37	18	1	1	0	15	0
11: Endothelial	50	3	23	20	21	26	9	6	2	50	23	560	32	17	28	13	9	0	3	3	0
12: Plasma cells	10	0	1	1	2	4	1	0	3	9	4	0	3	1	23	5	1	0	124	1	0
13: CAFs	555	25	71	21	7	32	12	4	1	28	11	12	12	28	65	21	4	0	0	3	0
14: Adipocytes	4	0	6	1	2	11	2	1	84	12	1	7	2	2	23	2	1	0	4	0	0

Table 6.3 — Confusion Matrix: 10x Native vs. Baysor (10x prior 1.0)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
0: Luminal epithelial	0	8	0	2	0	23	78	2612	0	12	4	1	0	10	6	8	0	0	0	8	1	0
1: Luminal epithelial	0	1	0	0	0	21	691	2052	0	13	4	1	0	6	1	73	0	1	0	5	0	1
2: Macrophages	4	203	0	36	8	6	25	3	1	7	4	1	3	3	3	1	2	9	1	5	1	0
3: Luminal epithelial	3	8	0	5	14	9	683	56	1	9	4	2	1	10	0	59	0	0	2	6	1	0
4: Myoepithelial	2	17	4	2	15	8	324	87	6	14	5	1	2	2	4	26	0	1	8	2	3	474
5: T cells	123	72	0	2335	49	4	5	0	40	25	9	4	18	5	13	0	26	23	6	16	23	0
6: B cells	25	27	0	34	18	5	1	0	10	9	4	1	5	2	5	0	3	20	8	3	556	0
7: Macrophages	108	2006	2	51	50	7	3	1	38	36	9	11	23	18	10	0	24	54	15	20	21	1
8: Luminal epithelial	12	6	1	1	7	13	310	123	4	12	5	1	2	4	3	1363	4	1	2	7	1	4
9: CAFs	2066	181	2	39	415	4	0	0	62	38	38	6	67	17	9	2	90	35	23	29	18	1
10: Smooth muscle	23	28	39	10	41	3	0	0	18	16	10	1	8	13	5	0	7	5	585	8	2	4
11: Endothelial	20	29	1125	46	37	5	2	0	11	32	11	4	10	14	3	2	4	8	71	15	6	2
12: Plasma cells	24	8	0	3	16	0	1	0	2	3	3	2	2	0	0	0	2	182	1	2	8	0
13: CAFs	78	113	6	27	863	2	24	7	16	12	17	1	19	1	5	12	4	15	24	10	7	2
14: Adipocytes	28	22	14	10	3	1	0	0	3	3	1	84	1	2	0	0	2	6	1	2	3	0

Table 6.4 — Confusion Matrix: 10x Native vs. Baysor (M prior 1.0)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
0: Luminal epithelial	8	3	0	144	0	0	0	39	2	0	0	0	9	0	0	0	5	2	0	2271	0	2	1
1: Luminal epithelial	1	3	0	1801	1	0	1	49	23	0	0	1	11	0	0	0	2	9	2	705	0	1	2
2: Macrophages	196	4	10	11	11	2	2	14	2	3	2	1	6	1	1	1	4	1	1	5	2	20	1
3: Luminal epithelial	10	2	7	647	13	3	1	26	54	3	0	2	17	1	0	2	2	1	0	11	0	1	4
4: Myoepithelial	12	4	1	311	4	1	0	19	6	2	1	5	5	6	1	3	12	3	7	17	1	1	486
5: T cells	81	17	2093	0	89	115	4	31	0	15	14	25	12	1	42	31	24	2	0	0	3	5	1
6: B cells	27	4	39	1	30	33	2	14	0	6	3	6	3	1	478	9	6	2	1	0	0	0	0
7: Macrophages	1789	51	53	3	94	98	9	40	1	12	10	29	13	7	23	38	23	4	6	0	2	0	2
8: Luminal epithelial	12	15	0	382	10	13	0	26	1173	2	5	2	7	1	0	6	5	4	0	39	0	0	4
9: CAFs	158	44	24	1	829	1652	3	28	0	27	37	56	25	6	18	79	18	8	18	0	5	0	1
10: Smooth muscle	22	8	9	0	33	14	0	9	1	4	1	18	4	91	3	19	12	7	476	0	1	0	3
11: Endothelial	31	8	42	1	40	14	2	14	1	12	2	8	15	1099	5	10	16	8	22	0	0	0	3
12: Plasma cells	9	1	5	0	19	16	0	2	0	3	0	1	1	1	188	1	3	0	1	0	0	0	0
13: CAFs	105	22	25	16	873	23	1	13	13	14	1	16	8	19	5	12	9	5	11	3	0	0	3
14: Adipocytes	20	3	14	0	4	27	80	5	0	0	0	3	2	9	8	2	0	0	1	0	0	0	0

Table 6.5 — Confusion Matrix: 10x Native vs. Baysor (CP prior 0.2)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
0: Luminal epithelial	0	23	4	1	14	2	0	2	0	0	4	4	2	9	0	4	0	0	0	0	0	1	0	245
1: Luminal epithelial	0	11	1	3	34	3	1	5	1	0	5	5	31	6	3	2	2	0	0	0	0	1	5	344
2: Macrophages	0	143	25	5	16	2	11	0	0	1	5	16	7	11	2	4	4	0	0	0	0	3	0	1
3: Luminal epithelial	4	15	7	6	15	8	37	1	0	5	9	4	133	5	3	1	4	2	0	0	0	0	0	15
4: Myoepithelial	0	21	5	14	11	7	17	4	5	0	24	5	2	5	1	2	2	0	0	0	3	1	319	28
5: T cells	48	44	1319	13	21	25	84	6	5	6	26	29	0	14	34	10	86	13	2	14	2	3	3	0
6: B cells	5	15	248	4	10	8	17	2	3	0	8	12	0	4	8	4	20	5	1	1	0	3	0	0
7: Macrophages	45	709	103	9	23	24	85	6	1	9	37	617	2	19	36	35	86	18	1	3	2	5	2	0
8: Luminal epithelial	18	17	0	4	22	9	19	3	3	0	15	0	289	9	15	2	12	7	0	0	1	1	4	37
9: CAFs	844	91	67	20	45	40	317	13	2	9	82	100	1	21	42	38	591	39	5	8	3	11	3	0
10: Smooth muscle	3	21	19	179	17	14	32	3	53	2	15	11	0	9	19	4	31	2	4	0	4	0	12	0
11: Endothelial	8	23	50	26	26	21	44	7	609	1	37	18	2	13	19	10	24	8	0	4	2	0	1	0
12: Plasma cells	6	2	10	1	1	1	17	0	2	1	3	4	0	2	2	2	17	1	0	141	0	0	0	0
13: CAFs	42	79	37	6	20	10	583	9	13	1	17	32	16	11	11	27	15	4	0	0	1	3	3	0
14: Adipocytes	3	8	13	0	3	0	7	1	10	79	2	12	0	4	0	2	20	0	0	4	0	1	0	0

Table 6.6 — Confusion Matrix: 10x Native vs. Baysor (CP prior 0.8)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
0: Luminal epithelial	0	6	6	0	38	2327	2	4	0	1	0	0	0	1	12	15	2	0	3	0
1: Luminal epithelial	0	4	2	1	489	1991	0	0	0	0	0	0	1	1	22	26	1	1	2	1
2: Macrophages	2	4	193	7	18	7	0	27	0	3	1	1	6	5	12	4	1	3	2	1
3: Luminal epithelial	3	4	10	16	590	80	1	7	0	3	3	3	1	3	14	12	2	2	1	0
4: Myoepithelial	3	1	20	14	244	109	4	5	4	0	5	0	2	21	8	6	4	1	5	423
5: T cells	132	7	67	59	2	2	11	1883	0	23	22	3	76	9	19	2	5	26	11	0
6: B cells	22	2	15	20	1	0	7	37	3	3	8	0	407	3	4	2	2	7	3	1
7: Macrophages	123	7	1746	69	2	2	22	61	1	25	29	11	56	19	32	1	4	27	5	1
8: Luminal epithelial	15	3	13	8	1458	120	5	0	1	5	1	0	1	2	4	9	0	9	2	4
9: CAFs	1664	15	185	536	3	0	39	36	7	68	33	4	52	32	34	5	15	76	14	4
10: Smooth muscle	21	4	27	38	1	1	4	9	31	6	13	0	5	523	10	0	1	15	7	5
11: Endothelial	22	15	34	35	4	0	5	47	974	7	16	1	20	72	19	2	6	13	13	1
12: Plasma cells	19	0	6	19	0	0	3	3	1	1	2	1	182	2	4	0	3	1	0	0
13: CAFs	58	2	118	772	28	3	5	28	10	5	12	1	23	24	15	2	7	8	11	1
14: Adipocytes	24	3	20	6	0	0	1	12	6	0	2	91	8	3	2	0	0	2	0	0

Table 6.7 — Confusion Matrix: 10x Native vs. Baysor (CP prior 1.0)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0: Luminal epithelial	0	7	0	0	5	102	0	0	1	1	1	16	0	0	2261	7	31	0	0	0	0
1: Luminal epithelial	0	1	1	1	2	1390	0	1	0	0	0	13	0	1	1058	41	54	0	1	0	2
2: Macrophages	2	182	0	10	1	7	2	4	4	27	3	12	0	3	6	2	22	2	2	2	1
3: Luminal epithelial	4	7	1	13	3	573	1	1	4	8	1	15	0	3	33	60	29	3	3	3	1
4: Myoepithelial	2	19	8	12	4	338	1	5	5	4	7	4	0	5	28	8	12	1	2	0	422
5: T cells	152	91	1	46	20	0	27	9	8	1922	5	9	7	40	0	0	22	6	13	5	0
6: B cells	30	22	1	16	6	0	3	5	4	433	7	2	3	11	0	0	7	0	2	1	0
7: Macrophages	138	1743	1	49	21	3	30	35	8	82	17	13	5	41	1	0	39	9	18	8	3
8: Luminal epithelial	13	10	2	6	8	355	5	4	4	0	1	5	0	7	44	1187	28	0	4	1	3
9: CAFs	1947	181	6	262	20	0	73	45	31	63	23	18	4	93	0	1	43	13	36	3	1
10: Smooth muscle	27	21	30	43	1	0	5	6	12	14	533	5	1	16	0	0	12	4	8	1	2
11: Endothelial	19	32	982	36	8	1	9	9	10	57	72	12	1	18	0	1	24	3	11	3	0
12: Plasma cells	20	9	0	14	2	0	2	2	1	9	2	1	170	3	0	0	3	3	0	1	0
13: CAFs	105	115	9	755	6	14	5	16	14	34	14	9	1	10	1	7	7	3	13	0	3
14: Adipocytes	25	19	10	3	1	0	0	1	0	14	3	5	5	4	0	0	4	2	1	84	0

Table 6.8 — Confusion Matrix: 10x Native vs. Baysor (SD prior 1.0)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
0: Luminal epithelial	0	0	1	27	72	6	18	4	3	0	0	29	2388	0	2	0	0	0	0	7	0	0
1: Luminal epithelial	1	0	1	29	1066	1	54	0	4	1	0	8	1491	0	11	0	0	0	1	7	0	3
2: Macrophages	9	4	3	13	17	193	4	38	2	3	0	7	3	1	2	1	0	1	7	2	1	0
3: Luminal epithelial	11	4	2	11	623	6	60	6	3	2	1	21	46	3	8	0	1	0	0	4	1	2
4: Myoepithelial	14	2	1	15	262	16	22	3	9	6	4	5	59	3	1	1	2	0	2	7	9	501
5: T cells	129	60	7	22	1	98	0	2225	12	32	0	12	0	34	2	29	7	13	17	17	7	0
6: B cells	37	14	4	11	1	28	0	32	1	4	0	5	0	10	1	532	6	2	17	3	4	1
7: Macrophages	139	48	11	45	3	1906	0	38	18	27	2	17	1	43	1	17	14	14	42	12	15	1
8: Luminal epithelial	18	8	1	14	278	6	1292	0	2	5	1	3	85	5	12	0	3	3	1	9	2	3
9: CAFs	1685	747	7	26	0	177	2	43	41	78	6	15	0	60	4	16	23	48	30	23	18	1
10: Smooth muscle	45	13	2	10	1	22	0	10	11	19	38	8	0	13	0	3	2	3	3	11	574	3
11: Endothelial	44	13	6	19	3	31	0	50	13	10	1102	10	0	7	0	6	7	8	4	9	63	4
12: Plasma cells	27	9	1	3	0	5	0	3	2	4	0	0	1	0	0	10	2	2	182	0	3	0
13: CAFs	874	15	3	11	19	104	15	22	21	29	10	5	2	21	3	8	7	3	19	10	17	1
14: Adipocytes	13	19	80	4	0	22	0	11	2	2	10	4	0	6	1	3	0	0	3	0	4	0

Table 6.9 — Confusion Matrix: 10x Native vs. CellPose

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12
0: Luminal epithelial	3	10	2	128	32	0	0	2253	0	1	1	0	0
1: Luminal epithelial	0	9	0	1332	77	0	0	1125	1	0	0	0	0
2: Macrophages	164	29	27	17	2	1	4	3	1	2	4	0	5
3: Luminal epithelial	2	22	3	577	99	0	4	23	2	0	2	0	0
4: Myoepithelial	12	12	3	261	22	0	8	59	451	1	10	5	2
5: T cells	55	73	1913	0	0	5	52	0	0	20	5	1	35
6: B cells	8	20	28	1	0	1	10	0	1	435	9	2	11
7: Macrophages	1705	101	53	1	0	12	82	0	3	29	27	1	49
8: Luminal epithelial	3	15	1	365	1220	0	3	29	2	3	0	0	1
9: CAFs	138	98	56	0	0	39	2038	0	7	32	26	4	19
10: Smooth muscle	14	26	8	0	0	3	33	0	10	3	577	11	2
11: Endothelial	23	21	42	3	0	1	26	0	5	4	114	978	7
12: Plasma cells	11	17	10	0	0	3	28	0	0	12	2	0	144
13: CAFs	81	47	30	11	10	28	743	0	6	9	32	7	17
14: Adipocytes	22	47	16	0	0	0	34	0	0	4	8	11	5

Table 6.10 — Confusion Matrix: 10x Native vs. Mesmer

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
0: Luminal epithelial	0	22	52	0	819	5	1537	1	0	0	0	0	41	0	1
1: Luminal epithelial	0	13	586	0	1784	0	137	9	0	0	0	0	49	0	1
2: Macrophages	8	34	17	1	2	183	2	6	5	2	1	3	1	0	21
3: Luminal epithelial	7	23	630	0	43	2	10	5	7	1	0	1	60	0	1
4: Myoepithelial	1	23	145	12	108	8	9	561	3	1	5	1	12	0	1
5: T cells	2307	84	1	1	0	65	0	1	14	32	1	40	0	2	7
6: B cells	19	24	2	4	0	13	0	0	13	568	2	7	0	1	0
7: Macrophages	51	161	1	9	0	1800	1	1	39	20	2	58	0	2	0
8: Luminal epithelial	1	21	441	1	25	1	19	9	4	0	0	2	1146	0	0
9: CAFs	57	116	1	16	0	169	0	3	744	20	2	1811	0	1	0
10: Smooth muscle	10	52	0	572	0	15	0	4	25	1	19	15	1	0	0
11: Endothelial	47	37	2	90	0	23	0	2	26	12	1069	9	1	0	0
12: Plasma cells	8	16	0	1	0	8	0	0	14	9	0	16	0	176	0
13: CAFs	19	77	11	15	3	104	0	7	820	10	11	74	16	0	0
14: Adipocytes	17	48	0	5	0	23	0	0	10	4	17	32	0	6	0

Table 6.11 — Confusion Matrix: 10x Native vs. StarDist

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11
0: Luminal epithelial	2373	109	0	47	3	2	0	1	5	1	1	24
1: Luminal epithelial	1739	821	1	74	0	1	0	1	0	1	5	10
2: Macrophages	9	16	9	6	184	32	1	11	14	2	2	15
3: Luminal epithelial	57	609	5	83	3	7	0	3	26	0	3	19
4: Myoepithelial	119	184	4	29	13	4	6	24	7	3	542	6
5: T cells	0	0	67	0	104	2356	1	15	79	27	1	36
6: B cells	0	1	15	0	29	34	2	16	22	582	0	18
7: Macrophages	0	0	112	1	1953	53	3	60	60	29	2	45
8: Luminal epithelial	60	402	8	1213	4	0	1	8	6	2	5	19
9: CAFs	0	0	2481	0	175	52	4	59	92	40	6	24
10: Smooth muscle	0	0	39	0	17	12	19	650	17	6	5	8
11: Endothelial	0	2	35	0	26	46	1098	137	12	7	5	11
12: Plasma cells	0	0	33	0	13	15	0	7	166	12	0	1
13: CAFs	2	16	886	16	106	27	7	48	65	11	7	13
14: Adipocytes	0	0	36	0	26	16	15	46	17	5	0	5

Table 6.12 — Confusion Matrix: 10x Native vs. Voronoi (CP)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0: Luminal epithelial	1	3	2257	0	154	0	0	2	0	0	13	0	0	0
1: Luminal epithelial	0	2	642	0	1822	0	3	52	0	0	23	0	0	0
2: Macrophages	17	176	21	0	3	0	10	13	2	10	6	1	0	0
3: Luminal epithelial	1	14	1	0	82	0	5	571	0	0	60	0	0	0
4: Myoepithelial	1	4	1	0	259	2	562	8	4	0	5	0	0	0
5: T cells	70	13	5	10	0	100	2	2	35	1892	0	14	0	16
6: B cells	12	7	2	6	0	21	4	1	33	69	0	0	0	371
7: Macrophages	1715	54	0	8	0	111	7	1	42	94	1	22	1	7
8: Luminal epithelial	0	71	17	0	335	1	3	19	2	1	1193	0	0	0
9: CAFs	98	54	0	3	0	2207	0	0	22	42	0	15	4	12
10: Smooth muscle	13	2	0	0	0	14	17	0	228	15	1	0	395	2
11: Endothelial	4	1	0	0	0	2	10	1	1081	10	0	2	113	0
12: Plasma cells	5	1	0	203	0	9	0	0	1	7	0	1	0	0
13: CAFs	89	399	0	5	0	349	10	13	108	36	6	0	0	6
14: Adipocytes	1	0	0	6	0	2	0	0	1	0	0	137	0	0

Table 6.13 — Confusion Matrix: 10x Native vs. Voronoi (10x)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0: Luminal epithelial	2717	0	42	0	0	8	0	0	0	0	0	0	17	0
1: Luminal epithelial	1223	0	1626	1	0	6	0	0	0	0	0	0	13	0
2: Macrophages	9	36	6	3	2	250	2	1	1	0	17	0	2	1
3: Luminal epithelial	11	0	622	6	0	238	0	0	0	0	3	0	17	0
4: Myoepithelial	8	1	321	681	0	3	4	0	0	0	2	0	3	0
5: T cells	0	133	2	1	1226	33	23	59	1309	4	10	0	0	6
6: B cells	1	8	1	4	644	15	22	21	19	3	5	0	0	1
7: Macrophages	0	2250	0	11	58	56	37	57	22	5	12	2	0	18
8: Luminal epithelial	30	0	469	8	0	157	0	0	0	0	3	0	1212	0
9: CAFs	0	139	0	0	1	3	17	2897	31	6	136	4	0	14
10: Smooth muscle	0	22	0	15	12	3	277	10	6	0	57	432	1	0
11: Endothelial	0	14	0	3	2	8	1424	5	0	1	6	3	0	3
12: Plasma cells	0	6	0	0	8	3	3	4	3	233	0	0	0	0
13: CAFs	0	112	0	25	2	53	25	292	32	5	756	1	1	2
14: Adipocytes	0	1	0	0	0	1	1	0	0	6	0	0	0	166

Table 6.14 — Confusion Matrix: 10x Native vs. Voronoi (M)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0: Luminal epithelial	0	98	0	0	0	0	0	3	0	0	14	0	4	2359
1: Luminal epithelial	0	2008	0	0	0	0	0	44	0	0	56	0	5	466
2: Macrophages	0	4	1	0	62	0	0	7	69	1	12	1	122	7
3: Luminal epithelial	0	499	0	0	1	0	0	6	22	1	245	3	12	1
4: Myoepithelial	0	129	0	0	1	1	0	729	0	0	18	8	0	4
5: T cells	3	0	0	60	111	26	10	2	30	2276	4	21	12	0
6: B cells	3	1	1	19	13	562	1	3	8	26	3	12	1	0
7: Macrophages	3	0	0	67	1920	9	21	5	29	40	0	27	24	0
8: Luminal epithelial	0	324	0	0	0	0	0	7	14	0	1307	0	0	18
9: CAFs	4	0	5	2566	105	5	17	0	196	29	0	13	0	0
10: Smooth muscle	0	0	377	5	14	4	0	6	17	11	1	279	0	0
11: Endothelial	0	1	1	5	8	4	7	2	6	16	1	1267	0	0
12: Plasma cells	218	0	0	7	5	8	1	0	2	3	0	4	0	0
13: CAFs	6	0	1	92	100	5	1	16	856	19	16	55	0	0
14: Adipocytes	5	0	0	0	1	0	153	0	0	0	1	2	0	0

Table 6.15 — Confusion Matrix: 10x Native vs. Voronoi (SD)

10x Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0: Luminal epithelial	2529	10	0	0	3	0	0	0	2	20	0	0	0	2	0	0
1: Luminal epithelial	1862	8	1	0	29	0	0	0	0	753	0	0	0	0	0	0
2: Macrophages	13	1	10	2	4	3	2	12	215	16	0	0	0	0	0	23
3: Luminal epithelial	16	24	14	0	14	1	1	0	9	735	0	1	0	0	0	0
4: Myoepithelial	41	4	5	0	771	0	8	0	6	104	0	1	0	1	0	0
5: T cells	0	1	19	62	2	2232	29	117	158	1	5	1	39	0	9	11
6: B cells	0	0	10	8	3	27	20	19	19	4	3	1	596	7	1	0
7: Macrophages	0	1	8	15	3	22	33	1111	1090	0	5	0	9	2	19	0
8: Luminal epithelial	33	1164	10	0	7	0	1	0	3	510	0	0	0	0	0	0
9: CAFs	0	1	201	2305	0	32	25	325	0	0	5	10	12	0	17	0
10: Smooth muscle	0	2	9	7	8	7	301	15	5	0	0	415	4	0	0	0
11: Endothelial	0	0	7	3	5	13	1323	4	7	1	0	4	6	1	5	0
12: Plasma cells	0	0	2	3	1	5	3	7	0	0	222	0	4	0	0	0
13: CAFs	0	21	802	55	19	25	68	178	14	12	5	1	4	0	0	0
14: Adipocytes	0	0	1	0	0	1	3	1	1	0	0	0	0	0	159	0

Section 7

Pseudobulk Correlation

Table 7.1 — Per-Cell-Type Pseudobulk Correlation vs. 10x Native

Method	ARI	Global r	Luminal epithelial	Macrophages	T cells	B cells	Myoepithelial	CAFs	Smooth muscle	Endothelial	Plasma cells	Adipocytes
CellPose	0.547	0.970	0.983	0.946	0.940	0.905	0.951	0.975	0.930	0.962	0.882	0.869
StarDist	0.545	0.975	0.985	0.952	0.944	0.927	0.965	0.979	0.930	0.967	0.908	0.884
Mesmer	0.557	0.983	0.989	0.970	0.972	0.959	0.974	0.985	0.972	0.980	0.935	0.920
Voronoi (CP)	0.630	1.000	1.000	0.997	0.997	0.984	0.989	0.999	0.982	0.998	0.984	0.976
Voronoi (SD)	0.584	1.000	1.000	0.998	0.999	0.993	0.996	0.999	0.990	0.999	0.986	0.984
Voronoi (M)	0.686	1.000	1.000	0.998	0.998	0.990	0.996	0.999	0.989	0.998	0.987	0.979
Baysor	0.305	0.999	0.997	0.973	0.980	0.958	0.970	0.997	0.976	0.994	0.937	0.949
Baysor (CP prior 0.8)	0.488	0.999	0.997	0.984	0.982	0.956	0.991	0.997	0.987	0.992	0.954	0.948
Baysor (CP prior 1.0)	0.501	0.999	0.997	0.985	0.983	0.973	0.993	0.997	0.988	0.993	0.953	0.946
Baysor (SD prior 1.0)	0.000	0.998	0.997	0.987	0.984	0.966	0.992	0.997	0.982	0.994	0.957	0.949
Baysor (M prior 1.0)	0.000	0.999	0.997	0.990	0.989	0.987	0.994	0.997	0.984	0.995	0.975	0.951

Gap: Missing 10x Ranger, Voronoi(10x), Baysor(CP prior 0.2), Baysor(10x prior 1.0). Baysor(SD/M prior 1.0) show ARI=0.0 (ARI not computed at generation time).

Table 7.2 — Per-Cluster Pseudobulk Correlation vs. 10x Native

method	0: Luminal epithelial	1: Luminal epithelial	2: Macrophages	3: Luminal epithelial	4: Myoepithelial	5: T cells	6: B cells	7: Macrophages	8: Luminal epithelial	9: CAFs	10: Smooth muscle	11: Endothelial	12: Plasma cells	13: CAFs	14: Adipocytes
CellPose	0.990	0.991	0.862	0.957	0.951	0.940	0.905	0.939	0.970	0.961	0.930	0.962	0.882	0.969	0.869
StarDist	0.991	0.992	0.873	0.957	0.965	0.944	0.927	0.952	0.972	0.974	0.930	0.967	0.908	0.972	0.884
Mesmer	0.994	0.995	0.914	0.974	0.974	0.972	0.959	0.965	0.982	0.981	0.972	0.980	0.935	0.984	0.920
Voronoi (CP)	0.998	0.999	0.969	0.996	0.989	0.997	0.984	0.996	0.997	0.998	0.982	0.998	0.984	0.996	0.976
Voronoi (SD)	0.999	0.999	0.981	0.994	0.996	0.999	0.993	0.998	0.998	0.999	0.990	0.999	0.986	0.998	0.984
Voronoi (M)	0.999	0.999	0.983	0.995	0.996	0.998	0.990	0.998	0.998	0.999	0.989	0.998	0.987	0.998	0.979
Baysor	0.995	0.993	0.882	0.987	0.970	0.980	0.958	0.977	0.995	0.997	0.976	0.994	0.937	0.990	0.949
Baysor (CP prior 0.8)	0.998	0.998	0.930	0.988	0.991	0.982	0.956	0.985	0.994	0.997	0.987	0.992	0.954	0.991	0.948
Baysor (CP prior 1.0)	0.999	0.999	0.932	0.989	0.993	0.983	0.973	0.987	0.993	0.997	0.988	0.993	0.953	0.991	0.946
Baysor (SD prior 1.0)	0.999	0.999	0.931	0.990	0.992	0.984	0.966	0.988	0.993	0.998	0.982	0.994	0.957	0.991	0.949
Baysor (M prior 1.0)	0.999	0.999	0.947	0.991	0.994	0.989	0.987	0.990	0.994	0.998	0.984	0.995	0.975	0.992	0.951

Section 8

Spatial Structure of Disagreement

Table 8.1 — Global Spatial Autocorrelation (Hungarian Alignment)

Comparison	n_cells	Disagree Rate	Moran's I	p_value
10x native vs. 10x native vs. ranger	23,155	35.0%	0.191	1.00e-04
10x native vs. baysor	10,953	51.7%	0.033	1.00e-04
10x native vs. baysor_10x native vs. ranger_prior_c10	22,910	34.7%	0.208	1.00e-04
10x native vs. baysor_mesmer_prior_c10	21,148	32.3%	0.115	1.00e-04
10x native vs. baysor_prior	11,454	51.9%	0.036	1.00e-04
10x native vs. baysor_prior_c08	20,108	38.8%	0.219	1.00e-04
10x native vs. baysor_prior_c10	20,308	33.8%	0.111	1.00e-04
10x native vs. baysor_stardist_prior_c10	21,814	37.7%	0.136	1.00e-04
10x native vs. cellpose	18,966	30.8%	0.178	1.00e-04
10x native vs. mesmer	20,595	27.9%	0.090	1.00e-04
10x native vs. stardist	21,429	33.5%	0.215	1.00e-04
10x native vs. voronoi	18,966	21.9%	0.076	1.00e-04
10x native vs. voronoi_10x native vs. ranger	23,153	28.3%	0.172	1.00e-04
10x native vs. voronoi_mesmer	20,595	18.8%	0.161	1.00e-04
10x native vs. voronoi_stardist	21,428	31.9%	0.194	1.00e-04

Table 8.2 — Global Spatial Autocorrelation (Argmax Alignment)

Comparison	n_cells	Disagree Rate	Moran's I	p_value
10x native vs. 10x native vs. ranger	23,155	34.5%	0.203	1.00e-04
10x native vs. baysor_10x native vs. ranger_prior_c10	22,910	33.1%	0.204	1.00e-04
10x native vs. baysor	10,953	43.8%	0.079	1.00e-04
10x native vs. baysor_mesmer_prior_c10	21,148	30.7%	0.119	1.00e-04
10x native vs. baysor_prior	11,454	39.2%	0.086	1.00e-04
10x native vs. baysor_prior_c08	20,108	37.4%	0.234	1.00e-04
10x native vs. baysor_prior_c10	20,308	32.1%	0.122	1.00e-04
10x native vs. baysor_stardist_prior_c10	21,814	32.9%	0.170	1.00e-04
10x native vs. cellpose	18,966	30.5%	0.189	1.00e-04
10x native vs. mesmer	20,595	27.3%	0.098	1.00e-04
10x native vs. stardist	21,429	33.4%	0.221	1.00e-04
10x native vs. voronoi_10x native vs. ranger	23,153	25.8%	0.168	1.00e-04
10x native vs. voronoi	18,966	21.9%	0.076	1.00e-04
10x native vs. voronoi_mesmer	20,595	18.8%	0.161	1.00e-04
10x native vs. voronoi_stardist	21,428	27.7%	0.229	1.00e-04

Table 8.3 — Spatial Autocorrelation (Non-10x Reference Pairs)

Comparison	n_cells	Disagree Rate	Moran's I	p_value
CellPose → Baysor (original)	8,947	48.8%	0.034	1.00e-04
cellpose → 10x	18,966	30.8%	0.176	1.00e-04
cellpose → baysor → prior	9,572	49.7%	0.051	1.00e-04
cellpose → stardist	19,460	13.8%	0.066	1.00e-04

Table 8.4 — LISA Cluster Distribution (Local Moran's I)

Comparison	n_cells	HH (hotspot)	HL	LH	LL (coldspot)
10x native vs. Baysor	10,953	2339 (21.4%)	3328 (30.4%)	3364 (30.7%)	1922 (17.5%)
10x native vs. Baysor (CP prior 0.2)	11,454	2532 (22.1%)	3412 (29.8%)	3497 (30.5%)	2013 (17.6%)
10x native vs. CellPose	18,966	4110 (21.7%)	1735 (9.1%)	7375 (38.9%)	5746 (30.3%)
10x native vs. Mesmer	20,595	3531 (17.1%)	2214 (10.8%)	8162 (39.6%)	6688 (32.5%)
10x native vs. StarDist	21,429	3984 (18.6%)	3191 (14.9%)	11047 (51.6%)	3207 (15.0%)
10x native vs. Voronoi (CP)	18,966	2106 (11.1%)	2055 (10.8%)	9650 (50.9%)	5155 (27.2%)
10x native vs. Voronoi (M)	20,595	1949 (9.5%)	1926 (9.4%)	12522 (60.8%)	4198 (20.4%)
10x native vs. Voronoi (SD)	21,428	4968 (23.2%)	1863 (8.7%)	7999 (37.3%)	6598 (30.8%)

Gap: LISA data for 8 of 17 comparisons. HH = disagreement hotspot, LL = agreement coldspot.

Section 9

Cell-Type Sensitivity

Table 9.1 — Per-Cell-Type Disagreement Rate (Pivoted)

Method	Adipocytes	B cells	CAFs	Endothelial	Luminal epithelial	Macrophages	Myoepithelial	Plasma cells	Smooth muscle	T cells
Baysor	49.1%	97.3%	47.3%	37.6%	57.5%	66.0%	36.2%	35.8%	94.7%	31.6%
CellPose	68.0%	17.3%	40.6%	20.1%	34.6%	26.6%	46.7%	36.6%	16.0%	11.4%
Mesmer	70.4%	13.0%	35.9%	18.9%	32.2%	25.1%	37.0%	29.0%	19.9%	9.7%
StarDist	100.0%	19.1%	40.0%	20.4%	43.0%	25.4%	42.4%	32.8%	15.9%	12.3%
Voronoi (CP)	6.8%	29.5%	25.1%	11.7%	20.5%	26.1%	33.6%	10.6%	42.5%	12.4%
Voronoi (M)	5.6%	13.9%	16.7%	3.9%	24.5%	16.0%	18.1%	12.1%	47.2%	10.9%
Voronoi (SD)	4.2%	17.0%	24.9%	4.1%	42.7%	49.4%	18.1%	10.1%	46.3%	16.9%

Gap: Only 7 methods. Missing: 10x Ranger, Voronoi(10x), all Baysor prior variants.

Table 9.2 — Cell-Type Disagreement (Full Detail) (page 1/3)

Method	Cell Type	n matched	n disagree	Disagree Rate
CellPose	Adipocytes	147	100	68.0%
CellPose	B cells	526	91	17.3%
CellPose	CAFs	3,478	1,412	40.6%
CellPose	Endothelial	1,224	246	20.1%
CellPose	Luminal epithelial	7,350	2,545	34.6%
CellPose	Macrophages	2,322	617	26.6%
CellPose	Myoepithelial	846	395	46.7%
CellPose	Plasma cells	227	83	36.6%
CellPose	Smooth muscle	687	110	16.0%
CellPose	T cells	2,159	246	11.4%
StarDist	Adipocytes	166	166	100.0%
StarDist	B cells	719	137	19.1%
StarDist	CAFs	4,137	1,656	40.0%
StarDist	Endothelial	1,379	281	20.4%
StarDist	Luminal epithelial	7,762	3,336	43.0%
StarDist	Macrophages	2,619	666	25.4%
StarDist	Myoepithelial	941	399	42.4%
StarDist	Plasma cells	247	81	32.8%
StarDist	Smooth muscle	773	123	15.9%
StarDist	T cells	2,686	330	12.3%
Mesmer	Adipocytes	162	114	70.4%
Mesmer	B cells	653	85	13.0%
Mesmer	CAFs	4,107	1,476	35.9%
Mesmer	Endothelial	1,318	249	18.9%
Mesmer	Luminal epithelial	7,517	2,420	32.2%

Table 9.2 — Cell-Type Disagreement (Full Detail) (page 2/3)

Method	Cell Type	n matched	n disagree	Disagree Rate
Mesmer	Macrophages	2,431	610	25.1%
Mesmer	Myoepithelial	890	329	37.0%
Mesmer	Plasma cells	248	72	29.0%
Mesmer	Smooth muscle	714	142	19.9%
Mesmer	T cells	2,555	248	9.7%
Voronoi (CP)	Adipocytes	147	10	6.8%
Voronoi (CP)	B cells	526	155	29.5%
Voronoi (CP)	CAFs	3,478	872	25.1%
Voronoi (CP)	Endothelial	1,224	143	11.7%
Voronoi (CP)	Luminal epithelial	7,350	1,507	20.5%
Voronoi (CP)	Macrophages	2,322	607	26.1%
Voronoi (CP)	Myoepithelial	846	284	33.6%
Voronoi (CP)	Plasma cells	227	24	10.6%
Voronoi (CP)	Smooth muscle	687	292	42.5%
Voronoi (CP)	T cells	2,159	267	12.4%
Voronoi (SD)	Adipocytes	166	7	4.2%
Voronoi (SD)	B cells	718	122	17.0%
Voronoi (SD)	CAFs	4,137	1,030	24.9%
Voronoi (SD)	Endothelial	1,379	56	4.1%
Voronoi (SD)	Luminal epithelial	7,762	3,316	42.7%
Voronoi (SD)	Macrophages	2,619	1,293	49.4%
Voronoi (SD)	Myoepithelial	941	170	18.1%
Voronoi (SD)	Plasma cells	247	25	10.1%
Voronoi (SD)	Smooth muscle	773	358	46.3%
Voronoi (SD)	T cells	2,686	454	16.9%

Table 9.2 — Cell-Type Disagreement (Full Detail) (page 3/3)

Method	Cell Type	n_matched	n_disagree	Disagree Rate
Voronoi (M)	Adipocytes	162	9	5.6%
Voronoi (M)	B cells	653	91	13.9%
Voronoi (M)	CAFs	4,107	685	16.7%
Voronoi (M)	Endothelial	1,318	51	3.9%
Voronoi (M)	Luminal epithelial	7,517	1,843	24.5%
Voronoi (M)	Macrophages	2,431	389	16.0%
Voronoi (M)	Myoepithelial	890	161	18.1%
Voronoi (M)	Plasma cells	248	30	12.1%
Voronoi (M)	Smooth muscle	714	337	47.2%
Voronoi (M)	T cells	2,555	279	10.9%
Baysor	Adipocytes	165	81	49.1%
Baysor	B cells	331	322	97.3%
Baysor	CAFs	3,262	1,544	47.3%
Baysor	Endothelial	898	338	37.6%
Baysor	Luminal epithelial	1,479	851	57.5%
Baysor	Macrophages	2,052	1,354	66.0%
Baysor	Myoepithelial	445	161	36.2%
Baysor	Plasma cells	193	69	35.8%
Baysor	Smooth muscle	436	413	94.7%
Baysor	T cells	1,692	534	31.6%

Section 10

Disagreement Drivers

Table 10.1 — Phenotypic Density vs. Disagreement

Comparison	n_agree	n_disagree	Med. Density (agree)	Med. Density (disagree)	U statistic	p_value
10x native vs. CellPose	13,121	5,845	-21.31	-20.78	42,184,301	2.93e-28
10x native vs. StarDist	14,254	7,175	-21.87	-20.63	59,767,412	1.06e-90
10x native vs. Mesmer	14,850	5,745	-21.73	-20.14	49,864,957	3.71e-79
10x native vs. Voronoi	14,805	4,161	-21.05	-21.35	30,394,953	0.1923
10x native vs. Voronoi (StarDist)	14,597	6,831	-21.74	-20.56	56,193,823	5.48e-51
10x native vs. Voronoi (Mesmer)	16,720	3,875	-21.38	-20.68	34,720,905	3.06e-12
10x native vs. Baysor	5,286	5,667	-22.76	-22.75	14,926,720	0.7570
10x native vs. Baysor (prior)	5,510	5,944	-22.99	-22.70	16,651,875	0.1183

Gap: 7 comparisons. Missing all Baysor PSC 0.8/1.0 variants, 10x Ranger, Voronoi(10x).

Table 10.2 — Cell Size vs. Disagreement (10x Native Cell Area)

Comparison	Median Area (agree)	Median Area (disagree)	p_value
10x native vs. CellPose	123.9 μm^2	121.4 μm^2	7.2e-07
10x native vs. StarDist	121.2 μm^2	116.9 μm^2	2.4e-12
10x native vs. Mesmer	126.2 μm^2	119.1 μm^2	3.2e-07
10x native vs. Voronoi (CP)	126.8 μm^2	111.5 μm^2	4.9e-32
10x native vs. Voronoi (SD)	123.8 μm^2	112.4 μm^2	8.8e-29
10x native vs. Voronoi (M)	125.9 μm^2	117.7 μm^2	3.7e-19
10x native vs. Baysor	167.0 μm^2	173.4 μm^2	0.28 n.s.

Table 10.3 — Top 20 DE Genes: 10x vs. baysor

Gene	Score	log2FC	adj p-value	% non-zero
FCER1G	11.69	0.61	5.25e-29	53.0%
CD14	10.39	0.92	5.25e-23	36.7%
ZEB2	9.88	0.45	6.40e-21	57.6%
MRC1	9.28	1.02	1.65e-18	26.6%
C1QC	9.04	0.92	1.16e-17	26.7%
CD163	8.53	1.00	9.58e-16	25.3%
AIF1	7.71	0.42	5.11e-13	44.1%
C1QA	7.59	0.88	1.18e-12	23.3%
CD68	7.36	0.59	6.02e-12	32.2%
CD4	6.64	0.42	9.48e-10	40.7%
CXCL16	6.38	0.49	4.92e-09	29.3%
FCGR3A	5.59	0.52	5.27e-07	22.8%
HAVCR2	5.31	0.59	2.21e-06	20.3%
CX3CR1	4.63	0.48	5.91e-05	21.4%
LYZ	4.44	0.10	1.32e-04	51.3%
ADAM9	4.41	0.25	1.49e-04	45.1%
FGL2	4.21	0.21	3.01e-04	33.1%
MS4A1	4.18	2.02	3.23e-04	7.4%
SMAP2	4.05	0.33	5.17e-04	26.7%
CCND1	3.75	0.15	0.0017	51.2%

Table 10.4 — Top 20 DE Genes: 10x vs. baysor_prior

Gene	Score	log2FC	adj p-value	% non-zero
LYZ	13.22	0.87	2.73e-37	52.9%
FCER1G	9.45	0.43	3.15e-19	52.7%
FGL2	8.44	0.70	1.97e-15	34.5%
SFRP4	7.91	0.34	1.35e-13	67.6%
CCDC80	7.56	0.27	1.92e-12	70.8%
FCER1A	7.09	1.46	5.69e-11	15.2%
ITGAX	7.07	1.20	6.03e-11	15.2%
AIF1	6.45	0.35	3.92e-09	43.5%
BASP1	6.09	0.40	3.54e-08	37.7%
LDHA	5.95	0.13	7.86e-08	82.7%
ZEB2	5.73	0.26	2.25e-07	56.4%
CD83	5.48	1.03	7.62e-07	14.2%
MNDA	5.27	0.69	1.95e-06	20.2%
CXCL16	5.20	0.45	2.70e-06	28.4%
BANK1	4.84	1.82	1.48e-05	8.9%
MS4A1	4.41	2.03	1.04e-04	7.8%
CX3CR1	4.21	0.48	2.40e-04	20.7%
IGSF6	4.15	0.47	3.05e-04	21.8%
FBLN1	4.09	0.19	3.82e-04	49.7%
LUM	3.83	0.16	9.61e-04	79.4%

Table 10.5 — Top 20 DE Genes: 10x vs. cellpose

Gene	Score	log2FC	adj p-value	% non-zero
SERPINA3	23.33	0.95	8.79e-118	60.4%
CLIC6	22.45	0.82	2.48e-109	64.3%
MUC1	22.38	0.87	7.97e-109	59.6%
MYBPC1	21.26	0.90	2.67e-98	55.3%
TPD52	21.04	0.75	2.20e-96	60.2%
PGR	20.73	0.88	1.26e-93	50.9%
FASN	20.21	0.94	4.45e-89	44.8%
GATA3	16.50	0.63	1.64e-59	66.0%
S100A14	16.43	0.66	5.14e-59	49.4%
KRT8	15.31	0.58	2.41e-51	52.0%
ENAH	14.97	0.47	4.10e-49	63.6%
RUNX1	14.76	0.40	8.80e-48	74.3%
TACSTD2	14.45	0.51	7.46e-46	55.8%
ERBB2	13.51	0.48	3.67e-40	54.0%
FOXA1	13.44	0.49	8.77e-40	50.4%
EPCAM	12.27	0.44	2.89e-33	50.1%
ANKRD30A	11.61	0.45	6.97e-30	56.8%
TMEM147	11.08	0.40	2.86e-27	50.9%
MLPH	10.27	0.38	1.61e-23	47.7%
POSTN	9.68	0.60	5.24e-21	46.6%

Table 10.6 — Top 20 DE Genes: 10x vs. mesmer

Gene	Score	log2FC	adj p-value	% non-zero
MYBPC1	18.66	0.83	3.64e-75	51.4%
SERPINA3	18.21	0.77	7.52e-72	55.7%
CLIC6	17.91	0.67	1.21e-69	60.1%
PGR	15.31	0.68	5.99e-51	46.2%
GATA3	14.72	0.57	3.70e-47	61.1%
FOXA1	14.69	0.61	4.67e-47	47.5%
MUC1	14.60	0.57	1.59e-46	54.3%
ANKRD30A	12.64	0.52	5.29e-35	53.8%
FASN	12.51	0.60	2.15e-34	39.2%
RUNX1	12.36	0.34	1.39e-33	72.1%
TPD52	11.78	0.42	1.37e-30	55.2%
EPCAM	11.17	0.45	1.49e-27	46.5%
MLPH	11.16	0.45	1.55e-27	45.7%
ELOVL5	11.14	0.39	1.72e-27	61.3%
ESR1	10.70	0.50	2.26e-25	40.4%
TACSTD2	10.19	0.36	4.49e-23	51.6%
ERBB2	9.96	0.37	4.39e-22	50.3%
KRT8	9.65	0.37	9.23e-21	47.7%
FLNB	9.54	0.33	2.34e-20	60.6%
S100A14	9.34	0.37	1.58e-19	43.8%

Table 10.7 — Top 20 DE Genes: 10x vs. stardist

Gene	Score	log2FC	adj p-value	% non-zero
CLIC6	34.01	1.19	5.72e-251	65.1%
MUC1	32.91	1.23	3.11e-235	60.3%
SERPINA3	32.18	1.23	4.10e-225	61.2%
MYBPC1	31.91	1.33	2.03e-221	56.0%
PGR	30.07	1.24	9.51e-197	51.8%
FASN	29.45	1.34	7.38e-189	46.3%
TPD52	28.85	0.97	2.78e-181	60.9%
GATA3	27.56	0.98	1.58e-165	66.6%
S100A14	23.93	0.92	6.29e-125	50.0%
TACSTD2	23.37	0.80	3.56e-119	56.5%
RUNX1	22.30	0.57	1.26e-108	73.9%
FOXA1	22.02	0.79	6.54e-106	51.3%
KRT8	21.99	0.79	1.10e-105	52.4%
ENAH	21.21	0.62	1.94e-98	63.8%
ERBB2	21.12	0.73	1.35e-97	54.4%
ANKRD30A	20.75	0.77	3.20e-94	57.7%
EPCAM	20.69	0.73	9.05e-94	51.2%
ELOVL5	19.73	0.65	2.35e-85	64.0%
MLPH	18.20	0.65	9.56e-73	48.8%
TMEM147	15.73	0.55	1.34e-54	50.6%

Table 10.8 — Top 20 DE Genes: 10x vs. voronoi

Gene	Score	log2FC	adj p-value	% non-zero
RUNX1	8.93	0.27	1.57e-16	71.4%
MYBPC1	8.07	0.42	8.86e-14	47.9%
MUC1	7.30	0.33	2.79e-11	52.6%
SERPINA3	7.24	0.33	3.45e-11	53.3%
CLIC6	7.20	0.31	3.71e-11	56.6%
PGR	5.83	0.30	2.56e-07	42.9%
POSTN	5.27	0.41	4.34e-06	44.7%
FASN	5.27	0.28	4.34e-06	36.8%
PDGFRB	4.81	0.60	3.80e-05	25.8%
DST	4.64	0.14	7.23e-05	45.2%
GATA3	4.46	0.20	1.51e-04	58.7%
TACSTD2	4.29	0.19	2.93e-04	49.5%
ANKRD30A	4.27	0.19	3.15e-04	51.4%
IGF1R	4.11	0.19	5.32e-04	44.3%
TPD52	4.08	0.17	5.78e-04	52.3%
TGFBR1	3.88	0.23	0.0012	46.1%
LARS	3.83	0.18	0.0014	48.4%
ELOVL5	3.67	0.16	0.0024	59.3%
ENAH	3.54	0.13	0.0038	58.5%
RGS5	3.50	0.70	0.0042	15.7%

Table 10.9 — Top 20 DE Genes: 10x vs. voronoi_mesmer

Gene	Score	log2FC	adj p-value	% non-zero
MUC1	20.28	0.92	7.36e-89	58.3%
SERPINA3	20.00	0.95	1.01e-86	59.2%
CLIC6	18.45	0.79	7.22e-74	62.3%
ENAH	17.46	0.64	2.37e-66	64.1%
TPD52	17.46	0.72	2.37e-66	58.9%
PGR	17.03	0.83	3.22e-63	50.0%
MYBPC1	16.02	0.77	5.45e-56	53.8%
FASN	15.81	0.82	1.36e-54	44.0%
GATA3	14.80	0.66	5.91e-48	63.7%
S100A14	14.34	0.70	4.97e-45	48.2%
KRT8	14.28	0.65	1.01e-44	51.7%
TACSTD2	14.18	0.59	3.65e-44	55.4%
FOXA1	13.09	0.57	1.13e-37	50.0%
EPCAM	13.08	0.58	1.19e-37	50.4%
ANKRD30A	11.80	0.54	9.13e-31	56.2%
ERBB2	11.40	0.47	8.85e-29	53.2%
RUNX1	10.45	0.32	2.71e-24	71.9%
CCND1	9.65	0.35	8.58e-21	60.5%
STAT3	9.56	0.32	1.99e-20	63.6%
MLPH	9.02	0.36	2.88e-18	47.6%

Table 10.10 — Top 20 DE Genes: 10x vs. voronoi_stardist

Gene	Score	log2FC	adj p-value	% non-zero
MUC1	27.03	1.08	2.20e-158	56.1%
MYBPC1	26.63	1.17	5.47e-154	52.9%
CLIC6	25.22	0.93	3.39e-138	59.1%
PGR	24.94	1.09	2.56e-135	48.7%
SERPINA3	24.29	1.00	1.64e-128	56.4%
FASN	24.29	1.16	1.64e-128	43.6%
TPD52	22.23	0.79	8.58e-108	56.9%
RUNX1	19.88	0.52	2.54e-86	72.7%
GATA3	19.51	0.73	3.25e-83	60.9%
TACSTD2	17.36	0.63	4.52e-66	52.8%
FOXA1	17.13	0.64	1.95e-64	48.3%
S100A14	16.86	0.68	1.93e-62	46.1%
ANKRD30A	16.77	0.68	8.63e-62	54.6%
ELOVL5	16.51	0.57	5.91e-60	61.7%
ERBB2	15.93	0.59	6.53e-56	51.1%
ENAH	15.89	0.49	1.10e-55	61.0%
EPCAM	15.89	0.59	1.11e-55	48.2%
KRT8	15.85	0.60	2.02e-55	49.0%
MLPH	15.03	0.56	6.41e-50	47.0%
TMEM147	13.43	0.48	4.72e-40	49.1%

Table 10.11 — DE Summary Across Comparisons

Comparison	n_sig_genes (adj p<0.05)	Top Gene	Top log2FC	Top Score
10x vs. baysor	83	FCER1G	0.61	11.69
10x vs. baysor_prior	102	LYZ	0.87	13.22
10x vs. cellpose	187	SERPINA3	0.95	23.33
10x vs. mesmer	155	MYBPC1	0.83	18.66
10x vs. stardist	231	CLIC6	1.19	34.01
10x vs. voronoi	71	RUNX1	0.27	8.93
10x vs. voronoi_mesmer	177	MUC1	0.92	20.28
10x vs. voronoi_stardist	232	MUC1	1.08	27.03

Gap: DE for 8 methods. Missing: Voronoi(10x), Baysor PSC 0.8/1.0 variants, 10x Ranger.

Section 11

Negative Marker Analysis

Table 11.1 — Negative Marker Summary

Method	n_cells	Med. tx/cell	Tier1 Violation Rate	Tier1 Violations	Mean Contam. Score	Violations / 1000 tx
10x native	23,629	124	4.5%	1,064	0.0052	0.36
CellPose	20,166	49	0.6%	119	0.0030	0.12
StarDist	24,745	45	0.6%	149	0.0031	0.13
Mesmer	21,697	70	1.4%	308	0.0040	0.20
Voronoi (CP)	20,166	149	6.4%	1,300	0.0054	0.43
Voronoi (SD)	24,743	122	4.5%	1,111	0.0050	0.37
Voronoi (M)	21,697	142	5.5%	1,189	0.0054	0.39
Baysor	18,321	53	5.0%	918	0.0034	0.95
Baysor (prior 0.8)	29,771	67	2.2%	664	0.0032	0.33
Baysor (prior 1.0)	30,473	69	2.2%	661	0.0033	0.31

Gap: 10 of 17 methods. Missing: Voronoi(10x), Baysor(SD/M/10x prior 1.0).

Table 11.2 — Per-Pair Violation Rates by Method

Marker Pair	10x native	Baysor	Baysor (prior 0.8)	Baysor (prior 1.0)	CellPose	Mesmer	StarDist	Voronoi (CP)	Voronoi (M)	Voronoi (SD)
ADIPOQ+CD3E	0.0014	4.91e-04	3.36e-04	2.63e-04	9.92e-05	1.84e-04	0.00e+00	0.0023	0.0018	0.0017
ADIPOQ+GATA3	0.0015	8.19e-04	5.04e-04	4.59e-04	4.96e-05	2.30e-04	4.04e-05	0.0022	0.0019	0.0019
CD14+ESR1	0.0020	0.0056	0.0010	0.0011	2.98e-04	5.07e-04	8.08e-05	0.0029	0.0024	0.0019
CD3E+ESR1	5.92e-04	0.0021	3.69e-04	5.25e-04	4.96e-05	5.53e-04	1.62e-04	0.0015	0.0011	6.47e-04
CD3E+GATA3	0.0097	0.0147	0.0059	0.0059	0.0020	0.0053	0.0021	0.0160	0.0122	0.0097
LUM+CD3E	0.0574	0.0409	0.0229	0.0242	0.0073	0.0230	0.0085	0.0746	0.0648	0.0538
LYZ+GATA3	0.0308	0.0352	0.0154	0.0149	0.0031	0.0073	0.0033	0.0436	0.0373	0.0304
MS4A1+GATA3	3.39e-04	0.0031	7.05e-04	5.25e-04	0.00e+00	0.00e+00	0.00e+00	0.0015	6.91e-04	6.47e-04
MS4A1+KRT14	8.46e-05	1.09e-04	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00	2.48e-04	1.38e-04	1.21e-04
MZB1+GATA3	5.50e-04	0.0017	3.69e-04	4.27e-04	9.92e-05	9.22e-05	8.08e-05	0.0011	5.99e-04	6.06e-04
PECAM1+ESR1	0.0014	0.0041	8.06e-04	7.55e-04	4.96e-05	2.30e-04	1.21e-04	0.0019	0.0017	0.0012
TRAC+KRT14	2.96e-04	8.19e-04	1.68e-04	9.84e-05	0.00e+00	9.22e-05	0.00e+00	5.45e-04	2.77e-04	2.02e-04
VWF+CD3E	0.0033	0.0020	9.41e-04	9.52e-04	4.46e-04	0.0015	4.45e-04	0.0057	0.0041	0.0035

Table 11.3 — Negative Marker Pair Definitions

Gene A	Gene B	Cell Type A	Cell Type B	Tier
CD3E	GATA3	T cell	Luminal epithelial	1
CD3E	ESR1	T cell	Luminal epithelial	1
TRAC	KRT14	T cell	Myoepithelial	1
MS4A1	GATA3	B cell	Luminal epithelial	1
MS4A1	KRT14	B cell	Myoepithelial	1
PECAM1	ESR1	Endothelial	Luminal epithelial	1
VWF	CD3E	Endothelial	T cell	1
CD14	ESR1	Macrophage	Luminal epithelial	1
LYZ	GATA3	Macrophage	Luminal epithelial	1
ADIPOQ	CD3E	Adipocyte	T cell	1
ADIPOQ	GATA3	Adipocyte	Luminal epithelial	1
LUM	CD3E	CAF	T cell	2
MZB1	GATA3	Plasma cell	Luminal epithelial	2

Section 12

Marker Gene Recovery

Gap Note

No CSV table exists for marker gene recovery.

Data is computed on-the-fly from AnnData objects in `scripts/run_marker_recovery.py` and exists only as a figure (`results/figures/marker_recovery.png`).

Re-run the marker recovery script with CSV output to include this as a table.

Section 13

Gaps & Coverage Matrix

Table 13.1 — Analysis Coverage Matrix

Method	Family	Cell Counts	Expr. Corr.	ARI	Confusion Matrix	Pseudobulk	Res. Sensitivity	Cell-Type Disagree	Density Disagree	DE Analysis	Neg. Marker	LISA
10x native	Reference	✓	—	—	—	—	—	—	✓	—	✓	—
CellPose	Nuclear	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
StarDist	Nuclear	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mesmer	Nuclear	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
10x Ranger	Nuclear	✓	✓	—	✓	—	—	—	—	—	—	—
Voronoi (CP)	Voronoi	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	✓
Voronoi (SD)	Voronoi	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	✓
Voronoi (M)	Voronoi	✓	✓	✓	✓	✓	✓	✓	—	✓	✓	✓
Voronoi (10x)	Voronoi	✓	✓	—	✓	—	—	—	—	—	—	—
Baysor	Transcript-density	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Baysor (CP prior 0.2)	Transcript-density	✓	✓	—	✓	—	—	—	—	✓	—	✓
Baysor (CP prior 0.8)	Transcript-density	✓	✓	—	✓	✓	✓	—	—	—	✓	—
Baysor (CP prior 1.0)	Transcript-density	✓	✓	—	✓	✓	✓	—	—	—	✓	—
Baysor (SD prior 1.0)	Transcript-density	✓	✓	—	✓	✓	✓	—	—	—	—	—
Baysor (M prior 1.0)	Transcript-density	✓	✓	—	✓	✓	✓	—	—	—	—	—
Baysor (10x prior 1.0)	Transcript-density	✓	✓	—	✓	—	—	—	—	—	—	—

Table 13.2 — Known Data Issues & Gaps

Issue	Status	Detail
Voronoi(10x) median expression correlation	Pending	Gene-name encoding mismatch noted as fixed; needs re-run
Baysor PSC sweep limited to CellPose prior	By design	Only CellPose tested at PSC 0.2/0.8/1.0; other detectors only at 1.0
LISA limited to 8 comparisons	Partial	Missing Baysor PSC 0.8/1.0, 10x Ranger, Voronoi(10x)
Cell-type disagreement limited to 7 methods	Partial	Missing all Baysor prior variants, 10x Ranger, Voronoi(10x)
DE analysis limited to 8 methods	Partial	Missing Voronoi(10x), Baysor PSC 0.8/1.0, 10x Ranger
Negative marker limited to 10 methods	Partial	Missing Voronoi(10x), Baysor(SD/M/10x prior 1.0)
Pseudobulk ARI=0.0 for Baysor(SD/M prior 1.0)	Data issue	ARI not computed at pseudobulk generation time
Marker gene recovery	Figure only	No CSV output; data computed on-the-fly from AnnData